

Soln: $G(x)$: x eats granola every day $H(x)$: x is healthy

Domain: Set of all humans in the world.

Premise: 1. $\forall x (G(x) \Rightarrow H(x))$ 2. $\neg H(\text{Linda})$

Conclusion: $\neg G(\text{Linda})$

<u>Steps</u>	<u>Reason</u>
1. $\forall x (G(x) \Rightarrow H(x))$	Premise
2. $\neg H(\text{Linda})$	Premise
3. $G(\text{Linda}) \Rightarrow H(\text{Linda})$	Universal instantiation on 1
4. $\neg G(\text{Linda})$	Modus tollens on 3, 2

This is a valid argument since the conclusion is derived using valid rules of inference.

17) What is wrong with this argument? Let $H(x)$ be " x is happy." Given the premise $\exists x H(x)$, we conclude that $H(\text{Lola})$. Therefore, Lola is happy.

Soln: We have the premise $\exists x H(x)$. By existential instantiation, for some element c in the set of all people, $H(c)$ is True, i.e. c is happy. But it's not necessarily true that c is Lola, c can be anybody else. $\therefore$ We cannot conclude $H(\text{Lola})$. $\square$

18) What is wrong with this argument? Let $S(x,y)$ be " x is shorter than y ." Given the premise $\exists s S(s, \text{Max})$, it follows that $S(\text{Max}, \text{Max})$. Then by existential generalization it follows that $\exists x S(x,x)$, so that someone is shorter than himself.

Soln: ~~We have the premise $\exists s S(s, \text{Max})$.~~ We have the premise $\exists s S(s, \text{Max})$. $\therefore$ For some person c in the domain, c is shorter than Max. Any person is not shorter than himself/herself. $\therefore \exists x S(x,x)$ is False by defn. of existential quantifier. $\therefore S(\text{Max}, \text{Max})$ is False. $\therefore s$ cannot be Max. $\therefore S(\text{Max}, \text{Max})$ does not follow from $\exists s S(s, \text{Max})$.